

LIN7076 – Foundations of Computational Linguistics

Regular Expressions and Automata

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Deep intuitions about well-formedness

- (1) a. Colorless green ideas sleep furiously.
- b. * Furiously colorless green sleep ideas.

'Twas brillig, and the slithy toves
Did gyre and gimble in the wabe:
All mimsy were the borogoves,
And the mome raths outgrabe.
“Beware the Jabberwock, my son!
The jaws that bite, the claws that catch!
Beware the Jubjub bird, and shun
The frumious Bandersnatch!”

Jabberwocky, Lewis Carroll

Can we come up with simple rules summarizing a language or a fragment thereof?

- We'll take a break from statistics and algebra to talk a bit about formalizing language structure.
- We'll explore two main approaches: **regular expressions** (this week), **context-free grammars** (next week).
- We'll discuss how these can be used for concrete applications: find-and-replace, parsing and structure-mining.
- We'll also review some aspects of natural language that are fundamentally challenging for these approaches.

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Examples of ungrammaticality/ill-formedness

- Tensed embedded clause cannot be subject-less (“ban on hyperraising”):

(2) a. * Jo seems [that won]. b. Jo seems [to have won].

- Subjects and main verbs must agree:

(3) a. * The dogs barks. b. The dog barks.

- Adverbs must precede main verbs:

(4) a. * Jo eats often apples. b. Jo often eats apples.

- All these generalizations are independent of meaning—they are **syntactic**, and sometimes language-dependent.
- Violating them causes ungrammaticality/**ill-formedness**.

Examples of ungrammaticality/ill-formedness

- Tensed embedded clause cannot be subject-less (“ban on hyperraising”):

(5) a. * Jo seems [that won]. b. Jo seems [to have won].

- Subjects and main verbs must agree:

(6) a. * The dogs barks. b. The dog barks.

- Adverbs must precede main verbs:

(7) a. * Jo eats often apples. b. Jo often eats apples.

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- Tensed embedded clause cannot be subject-less (“ban on hyperraising”):

(8) a. * Jo seems [that won]. b. Jo seems [to have won].

- Subjects and main verbs must agree:

(9) a. * The dogs barks. b. The dog barks.

- Adverbs must precede main verbs:

(10) a. * Jo eats often apples. b. Jo often eats apples.

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- Tensed embedded clause cannot be subject-less (“ban on hyperraising”):

(11) a. * Jo seems [that won]. b. Jo seems [to have won].

- Subjects and main verbs must agree:

(12) a. * The dogs barks. b. The dog barks.

- Adverbs must precede main verbs:

(13) a. * Jo eats often apples. b. Jo often eats apples.

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- Tensed embedded clause cannot be subject-less (“ban on hyperraising”):

(14) a. * Jo seems [that won]. b. Jo seems [to have won].

- Subjects and main verbs must agree:

(15) a. * The dogs barks. b. The dog barks.

- Adverbs must precede main verbs:

(16) a. * Jo eats often apples. b. Jo often eats apples.

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Examples of infelicity

- A proposition and its negation cannot be conjoined (contradictory!) or disjoined (tautology!):

(17) a. # Jo is inside and outside. b. # Jo is inside or outside.

- Sentences should take obvious facts for granted...

(18) a. # A sun is shining. b. The sun is shining.
 ↪ There's at least one sun. ↪ There's a unique sun.

- But should not take new information for granted:

(19) a. ?? Jo will need to bring her alligator to the vet. b. Jo has an alligator, and will need to bring it to the vet.

- These generalizations depend on meaning (semantic) and largely language-independent. Violating them causes infelicity.

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- A proposition and its negation cannot be conjoined (contradictory!) or disjoined (tautology!):

(20) a. # Jo is inside and outside. b. # Jo is inside or outside.

- Sentences should take obvious facts for granted...

(21) a. # A sun is shining. b. The sun is shining.
 $\rightsquigarrow$ There's at least one sun. $\rightsquigarrow$ There's a unique sun.

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(23) a. # Jo is inside and outside. b. # Jo is inside or outside.

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(24) a. # A sun is shining. b. The sun is shining.
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- But should not take new information for granted:

(25) a. ?? Jo will need to bring **her**
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- A proposition and its negation cannot be conjoined (contradictory!) or disjoined (tautology!):

(26) a. # Jo is inside and outside. b. # Jo is inside or outside.

- Sentences should take obvious facts for granted...

(27) a. # A sun is shining. b. The sun is shining.
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(28) a. ?? Jo will need to bring **her**
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- A proposition and its negation cannot be conjoined (contradictory!) or disjoined (tautology!):

(29) a. #Jo is inside and outside. b. #Jo is inside or outside.

- Sentences should take obvious facts for granted...

(30) a. # A sun is shining.
 $\rightsquigarrow$ There's at least one sun.
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- But should not take new information for granted:

(31) a. ??Jo will need to bring **her**
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(33) a. # A sun is shining.
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Formalizing well-formedness

- Today we'll **focus on modeling grammaticality/well-formedness**.
- A language can be modeled as the set of all the sentences that are well-formed in that language—independently of meaning.
- For instance, English = {The dog barks, Colorless green ideas sleep furiously, ...}. How big is this set?

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Other possible specifications of a “language”

- More abstractly, a language can be seen as the set of well-formed sequences of syntactic categories (instead of words instantiating these categories).
- In that case, English = {Det Noun Verb, Adj Adj Noun Verb Adv, ...}. How big is this set?
- More narrowly, a language can be seen as the set of well-formed sequences of phonological elements (e.g. consonants C, and vowels V¹)
- English is hard to illustrate, so let's take Japanese whose building blocks are mostly CV syllables plus occasional nasals. In that case Japanese = {CV, CVCV, CVCVCV, CVnCV, CVCVn, ...}.
- In all these cases, we are interested in **well-formed units of a language** (strings of words; strings of syntactic labels; strings of phonemes...).

¹One can choose more specific categories too!

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The need for recursivity

- To model natural languages, formal languages need to generate **an infinite number of outputs from finite means**.
- Finite means:
 - A finite alphabet Σ of primitive “words” (lexical items, or syntactic categories).
 - A finite set of rules with some degree of recursivity.
- Recursivity is the **capacity for a rule to refer to itself**. Because recursive rules may apply an arbitrary number of times, an infinity of outputs can be generated.
- Concrete example: a folder-based file system is inherently recursive. Any **folder** may contain only **files** (“base case”), or, at least one other **folder** (“recursive case”).
- More generally, anything that displays an **arborescent structure** will be recursive!

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A toy example: intensified adjectival phrases

- Suppose our alphabet Σ is { a, the, very, tall, girl, boy }. What kind of well-formed phrases can we build out of this alphabet?
- *the boy* and *the girl* are well-formed.
- *a boy* and *a girl* are well-formed.
- *the/a tall boy/girl* are well-formed.
- *the/a very tall boy/girl* are well-formed.
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Regular expressions

- Let Σ be an alphabet. We call the members of Σ **words** (they can be “real” words like *cat*, syntactic labels like *Det*, phonological units like *V...*).
- Regular expressions (**regexes**) are used to specify **word patterns**.
- Regexes are built out of consist of **constants**, which denote sets of words, and **operator symbols**, which denote operations over these sets.
- Constants: $\emptyset$ (the empty regex), and $\{\epsilon\}$ (the empty-word regex) are regexes. For any w in Σ , $\{w\}$ is a regex.
- Operations: given R and S two regexes...
 - RS is the pairwise **concatenation** of R and S and is a regex.
 - $R|S$ is the **union** of R and S and is a regex.
 - R^* is the set containing ϵ and all possible concatenations of elements of R (“**Kleene star**”) and is a regex.
- We’ll use parentheses to signal the order of precedence between these operations.

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- The phrase ends with *boy* or *girl*. Achieved again with the **union** operation (*boy|girl*).
- **Concatenating** all these parts together form the desired pattern: (*a|the*)(*very**)*tall|* ϵ)(*boy|girl*).
- What is the issue with the regex (*a|the*)(*very**)(*tall|* ϵ)(*boy|girl*)?

Useful shorthands

- One can define useful **shorthands** from the three core operations (concatenation, union, Kleene star). If R and S are regexes:
 - R^+ is defined as RR^* : it is the set all possible concatenations of elements of R .
 - R^k with $k > 0$ a finite integer is defined as $\underbrace{RRR\dots R}_{k \text{ times}}$ is the set of all possible concatenations of elements of R involving exactly k elements.
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- Mining a dataset for specific verbs and all their **possible inflections** (including noisy/erroneous ones!), for instance **CHILDES** when studying the acquisition of verbal predicates.
- Mining a naturalistic corpus for a wide range of **contracted/colloquial forms**; for instance when studying language change or sociolinguistic variation.
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Graphical representation: automata

- Regexes can be represented using diagrams called **automata**.
- Automata have **states**, represented with **circles**. There's one **initial** state (where you start), and at least one **accepting** state (where you want to end, signaled with a double circle).
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- More generally, regex **concatenation** is represented by **arrows between states**.

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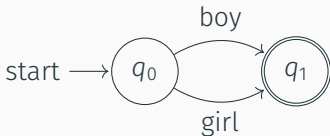
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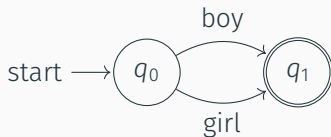
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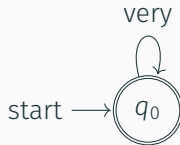
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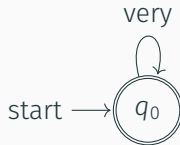
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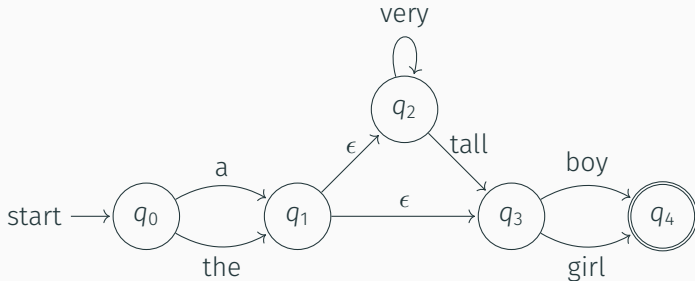
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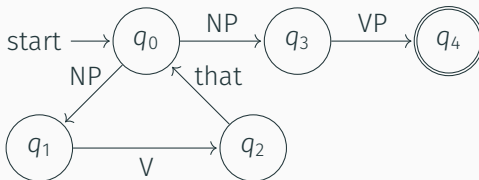
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Practice: automaton for right-embedded complementation

- Recall that our pattern for *Jo believes that Al thinks... that Ed is happy...* was: $(NP\ V\ that)^*(NP\ VP)$.
- It meant that first we want NP V that (an embedding sentence) zero or more times, and then we finish by NP VP (the last embedded sentence).
- What's the automaton for this?



Linguistic challenges for regular expressions and automata

Structural ambiguities

- Compare:

- (35) a. Tall boys and girls. c. Tall boys and short girls.
 b. Tall boys and tall girls. d. Boys and tall girls.

- When an adjective precedes conjoined NPs, as in (35a), it's **ambiguous whether the adjective applies to both NPs** (as in (35b)), **or to the first NP only** (with a meaning compatible with (35c)).
- This ambiguity disappears if the adjective follows the first and precedes the second NP, as in (35d).
- Similar-ish ambiguities arise in sentences like (36) and (37).

(36) The photographer saw the bird on the hill.

(37) They are building blocks.

Structural ambiguities

- Compare:

- (38) a. Tall boys and girls. c. Tall boys and short girls.
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(40) They are building blocks.

Structural ambiguities

- Compare:

- (41) a. Tall boys and girls. c. Tall boys and short girls.
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(42) The photographer saw the bird on the hill.

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Structural ambiguities

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- (44) a. Tall boys and girls. c. Tall boys and short girls.
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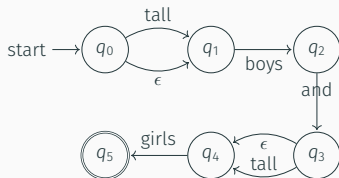
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Regexes/automata have a “weak” generative capacity

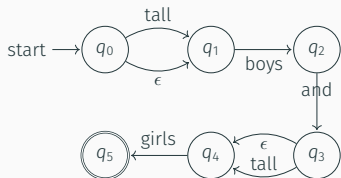
- Regexes/automata *can* match strings like *tall boys and girls*, *tall boys and tall girls* etc: $(\text{tall}|\epsilon)$ boys and $(\text{tall}|\epsilon)$ girls.



- But nothing in the traversal of the automata, or in the denotation of the regex (a sets of strings!) tells us that *tall boys and girls* can be understood either as *[tall [boys]] and [girls]* or as *[tall [boys and girls]]*!
- These devices capture the well-formedness of the string, but **nothing about its internal structure** (which influences meaning).
- In that sense, regexes and automata have a **weak generative capacity** w.r.t. structural ambiguities.

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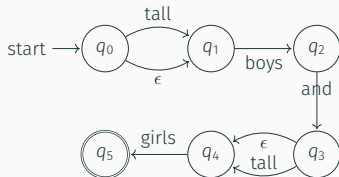
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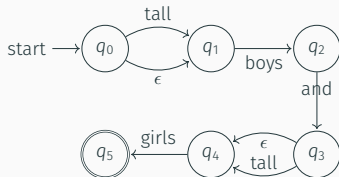
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Deeper challenge: center-embedding

- Let's consider nested subject-modifying, object-gap relative clauses:

- (47)
- a. The mouse is scared.
 - b. The mouse [that the cat chases] is scared.
 - c. The mouse [that the cat [that the dog barks at] chases] is scared.
 - d. The mouse [that the cat [that the dog [that the postman avoids] barks at] chases] is scared.
- The corresponding structures are: NP VP; NP that NP V VP; NP that NP that NP V V VP; NP that NP that NP that NP V V V VP... etc.
 - Can we synthesize this using one single regex?
 - The pattern appears to be: NP (that NP)^k (V)^k VP, **for any** $k \geq 0$. But there are an infinity of such k , so our “regex” would need to be infinitely long... Not a regex!

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Challenges summary

- The fundamental limitation of regexes/automata is that they **cannot keep track of arbitrary occurrence counts**. Anything that looks like a sequence of k identical patterns, followed by a sequence of k (or $k - 1$, or $k + 6...$) other patterns, cannot be matched by regexes/automata. They are **not expressive enough**.
- This sheds light on a fundamental complexity difference between right-embedding (*Jo thinks that Ed believes that...*), which can be captured by regexes, and center-embedding (*The mouse that the cat that... chases is scared*), which cannot.
- We have also seen that regexes and automata cannot account for the internal structure of strings, and the meaning differences that may arise from structural ambiguities.
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